

FA $\Leftrightarrow$ RG $\Leftrightarrow$ RE

I) FA $\Leftrightarrow$ RG (team work)

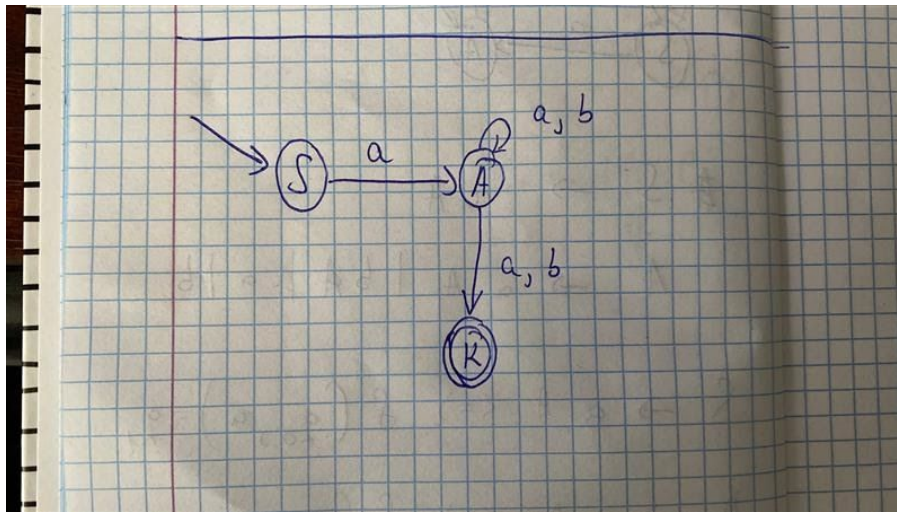
T1. Given the regular grammar $G = (\{S, A\}, \{a, b\}, P, S)$

$$P : S \rightarrow aA$$

$$A \rightarrow aA \mid bA \mid a \mid b,$$

build the equivalent FA.

Sol.:



T2. Given the regular grammar $G = (\{S, A\}, \{a, b\}, P, S)$

$$P : S \rightarrow \varepsilon \mid aA$$

$$A \rightarrow aA \mid bA \mid a \mid b,$$

build the equivalent FA.

Sol.:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{S, A, K\}, q_0 = S, F = \{K, S\}, \Sigma = \{a, b\}$$

δ	a	b
S	{A}	$\emptyset$
A	{A, K}	{A, K}
K	$\emptyset$	$\emptyset$

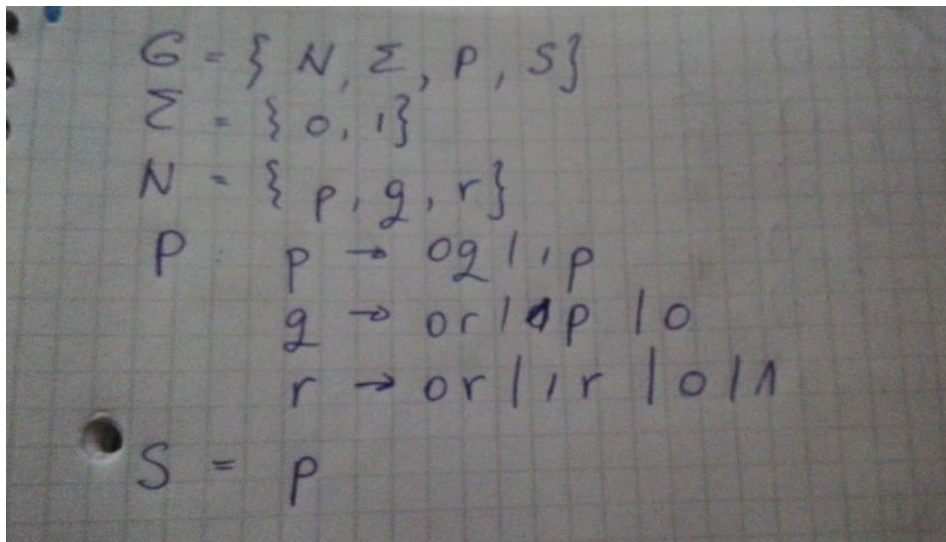
T3. Given the following FA $M = (Q, \Sigma, \delta, q_0, F)$

$$Q = \{p, q, r\}, q_0 = p, F = \{r\}, \Sigma = \{0, 1\}$$

δ	0	1
p	q	p
q	r	p
r	r	r

build the equivalent right linear grammar.

Sol.:



T4. Given the following FA $M = (Q, \Sigma, \delta, q_0, F)$

$$Q = \{p, q, r\}, q_0 = p, F = \{p, r\}, \Sigma = \{0, 1\}$$

δ	0	1
p	q	p
q	r	p
r	r	r

build the equivalent right linear grammar.

Sol.:

$$\begin{aligned}
 G &= \{N, \Sigma, P, S\} \\
 \Sigma &= \{0, 1\} \\
 N &= \{p, q, r\} \\
 S &= p \\
 P: \quad &p \rightarrow 0q \mid 1p \mid \epsilon \\
 &q \rightarrow 0r \mid 1p \mid 0 \mid 1 \\
 &r \rightarrow 0r \mid 1r \mid 0 \mid 1.
 \end{aligned}$$

II) RG $\Leftrightarrow$ RE

1. Give the RG corresponding to the following RE $0(0+1)^*1$.

$$0: G_1 = (\{S_1\}, \{0, 1\}, \{S_1 \rightarrow 0\}, S_1)$$

$$1: G_2 = (\{S_2\}, \{0, 1\}, \{S_2 \rightarrow 1\}, S_2)$$

$$0+1: G_3 = (\{S_1, S_2, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0, S_2 \rightarrow 1, S_3 \rightarrow 0 \mid 1\}, S_3)$$

$$G'_3 = (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow 0 \mid 1\}, S_3)$$

$$(0+1)^*: G_4 = (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow 0 \mid 1, S_3 \rightarrow 0S_3 \mid 1S_3, S_3 \rightarrow \epsilon\})$$

$$G'_4 = (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow 0S_3 \mid 1S_3 \mid \epsilon\}, S_3) \text{ ! not regular}$$

$$0(0+1)^*: G_5 = (\{S_1, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow 0S_3 \mid 1S_3 \mid \epsilon\}, S_1)$$

! not regular

$$0(0+1)^*1:$$

$$G_6 = (\{S_1, S_2, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow 0S_3 \mid 1S_3 \mid S_2, S_2 \rightarrow 1\}, S_1) \text{ ! not regular}$$

$$G'_6 = (\{S_1, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow 0S_3 \mid 1S_3 \mid 1\}, S_1)$$

(TW)

2. Give the RE corresponding to the following grammar

$$G = (\{S, A, B\}, \{a, b\}, P, S)$$

$$P : S \rightarrow aA$$

$$A \rightarrow aA \mid bB \mid b$$

$$B \rightarrow bB \mid b$$

Sol.: T4

Handwritten derivation of the regular expression for the given grammar:

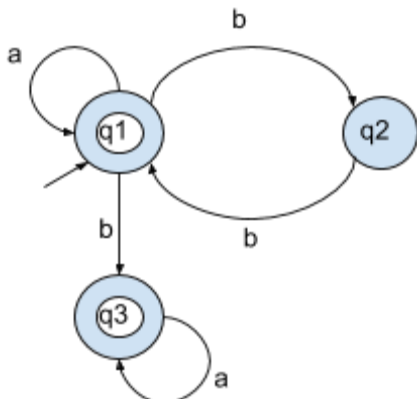
$$\begin{aligned} S &= aA \\ A &= aA + bB + b \\ B &= bB + b \Rightarrow B = b^*b = b^+ \\ \Rightarrow A &= a^* + B \\ A &= a^*B = a^*b^+ \\ S &= aA \Rightarrow S = aa^*b = a^+b^+ \end{aligned}$$

III) FA $\Leftrightarrow$ RE

1. Give the FA corresponding to the following RE $01(1+0)^*1^*$.

#board, pdf attached to Seminar 7 meet in MSTeams attached, at the end of the document

2. Give the regular expression corresponding to the FA below.



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$$q_1 = \varepsilon + q_1 a + q_2 b$$

$$q_2 = q_1 b$$

$$q_3 = q_1 b + q_3 a$$

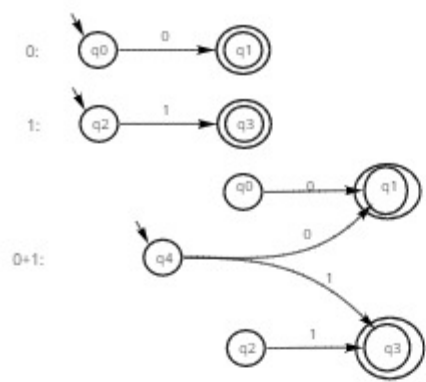
$$X = Xa + b \Rightarrow X = ba^* \text{ solution}$$

$$q_3 = q_1 ba^*$$

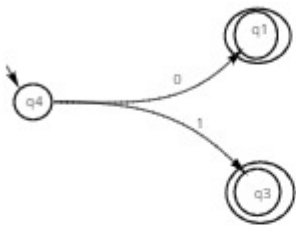
$$q_1 = \varepsilon + q_1 a + q_1 bb = q_1(a + bb) + \varepsilon \Rightarrow q_1 = (a + bb)^* \Rightarrow q_3 = (a + bb)^* ba^*$$

$$RE = q_1 + q_3 = (a + bb)^* + (a + bb)^* ba^* = (a + bb)^* (\varepsilon + ba^*)$$

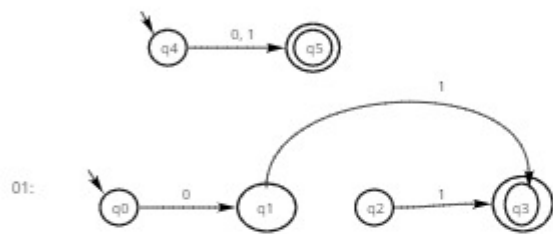
RE->FA



eliminate inaccessible



further simplify



eliminate inaccessible



(0+1)*:



simplify



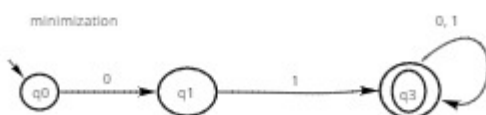
1*



01(0+1)*:



minimization



01(0+1)*1*:



equivalent to the previous one